Logarithms and Their Properties

Evaluating logarithms

1 Evaluate each logarithm exactly:

a $\log_2 32$ **b** $\log_3 \frac{1}{9}$ **c** $\log 1000$ **d** $\ln e^7$

2 Use the change-of-base formula to evaluate $\log_5 18$, to three decimal places.

3 State the domain of $f(x) = \log(x-4)$ and the inverse function of $g(x) = 2^x$.

Properties of logarithms

4 Expand fully using properties of logarithms: $\ln\left(\frac{x^3\sqrt{y}}{z^2}\right)$.

5 Write as a single logarithm: $2\log x + 3\log y - \log z$.

6 Given $\log_b 2 = 0.431$ and $\log_b 3 = 0.683$, evaluate:

a $\log_b 6$ **b** $\log_b 8$ **c** $\log_b 1.5$

Graphs of logarithmic functions

7 For $y = \log_2(x+4) - 1$, state:

a the domain **b** the vertical asymptote **c** the x-intercept

8 Describe how the graph of $g(x) = -\ln(x-2) + 3$ is obtained from $y = \ln x$.

Logarithmic scales and applications

9 The Richter scale measures earthquake magnitude by $M = \log\left(\frac{I}{I_0}\right)$, where I is intensity and I_0 is a reference intensity.

a An earthquake has $M = 6$. Express I in terms of I_0.

b How many times more intense is a magnitude-7 earthquake than a magnitude-5 earthquake?

10 Sound intensity level in decibels is $\beta = 10\log\left(\frac{I}{I_0}\right)$. If a sound's intensity doubles, by how much does β increase, to two decimal places?

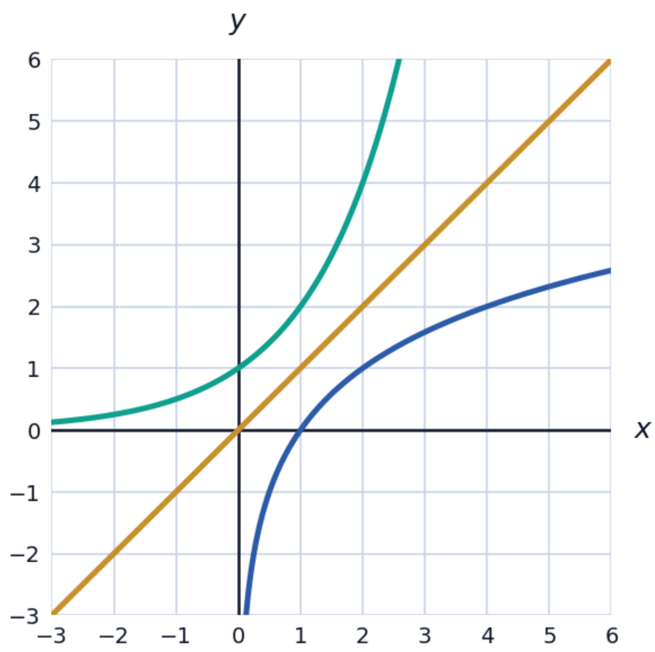
More practice with logarithm properties

11 Expand fully: $\log_2\left(\frac{8x^2}{\sqrt[3]{y}}\right)$.

12 Write as a single logarithm and simplify: $\frac{1}{2}\ln 16 + \ln 3 - \ln 2$.

Logarithms as inverse functions

13 The graphs of $y = 2^x$ and $y = \log_2 x$ are shown together with the line $y = x$. Explain why the two curves are reflections of each other across $y = x$.



Solving simple logarithmic equations

14 Solve $\log_3(x+2) = 4$ for x.

15 Solve $2\log x = \log(x+6) + \log 1$ for x (noting $\log 1 = 0$), and check for extraneous solutions.

The pH scale (a logarithmic application)

16 pH is defined as $pH = -\log[H^+]$, where $[H^+]$ is hydrogen ion concentration (mol/L). Find the pH of a solution with $[H^+] = 3.2 \times 10^{-5}$, to two decimal places.

17 Two solutions have pH values of 3 and 6. Find the ratio of their hydrogen-ion concentrations $[H^+]$, and state which is more acidic.

Common mistakes with logarithm properties

18 Explain why $\log(x+y) \neq \log x + \log y$ in general, using a specific numerical counterexample.

19 Simplify $\log_2(4x) - \log_2(2x)$ using the quotient property, and verify with a specific value of x.